2 **(a)** State the principle of moments.

....

....

.... [2]

(b) Three objects A, B and C are placed on a horizontal beam. The beam is in equilibrium, as shown in Fig. 2.1.

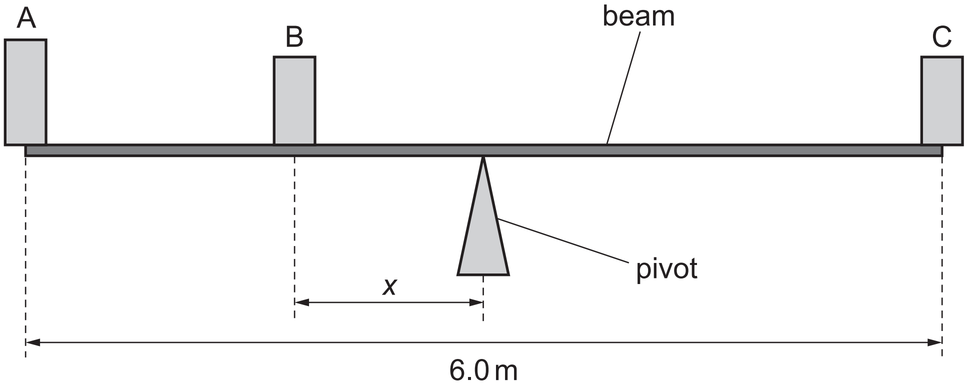

Fig. 2.1 (not to scale)

The beam is uniform and has length 6.0 m.
The pivot is at the midpoint of the beam.
Object A has mass 60 kg and is at one end of the beam.
Object B has mass 45 kg and is at a distance x from the pivot.
Object C has mass 80 kg and is at the other end of the beam.

Calculate x.

x = ... m [3]